Supply Chain Analytics Dashboard

Project Report

Power BI • DAX • Power Query • Supply Chain Analytics

Analyze • Visualize • Optimize

1. Project Overview

Attribute	Details
Project Title	Supply Chain Analytics Dashboard Using Power BI
Tool	Power BI Desktop
Data Source	DataCo Supply Chain Dataset (embedded in .pbix)
Report Pages	Two-page interactive report — Overview & Details
Domain	Supply Chain / Operations Analytics
Purpose	Sales performance, delivery efficiency, logistics analysis

2. Objective

The objective of this project is to analyze supply chain transactional data to derive meaningful insights related to sales performance, order trends, delivery efficiency, and geographic distribution. This dashboard supports data-driven operations and logistics decision-making by identifying patterns in delivery delays, customer segment behavior, and market-level revenue distribution.

3. Problem Statement

Organizations managing complex supply chains often struggle to answer:

- Which products and categories generate the most revenue?
- What percentage of orders are being delivered late?
- How do different shipping modes impact order volumes and delivery time?
- Which markets and customer segments drive the most sales?
- How have sales and order volumes trended over time?

This project addresses these challenges by consolidating supply chain KPIs and logistics data into a two-page interactive Power BI dashboard.

4. Dataset Description

The dashboard is powered by the DataCo Supply Chain Dataset — a publicly available supply chain simulation dataset containing transactional order data across multiple markets, product categories, and customer segments. No external database connection is required; all data is embedded within the .pbix file.

4.1 Key Fields

Field	Description
Order Id	Unique identifier for each order
Product Name	Name of the ordered product
Category Name	Product category grouping
Customer Segment	Segment: Consumer, Corporate, Home Office, etc.
Market	Geographic market region
Customer Country	Country of the customer
Shipping Mode	Method of shipment (Standard, First Class, etc.)
Delivery Status	On-time, Late, Advance, or Cancelled
Order Date	Date the order was placed

4.2 Calculated Measures (DAX)

Measure	Description
Total Sales	Sum of all revenue generated across orders
Total Orders	Count of all orders placed
Profit Margin %	Revenue minus cost expressed as a percentage
Avg Order Value	Mean revenue value per individual order
Late Delivery %	Percentage of orders delivered beyond expected date
Avg Delivery Time (Days)	Mean number of days from order placement to delivery

5. Tools & Technologies

Tool / Technology	Purpose
Power BI Desktop	Dashboard design, layout, and report publishing
Power Query (M Language)	Data ingestion, cleaning, and transformation
DAX (Data Analysis Expressions)	Custom KPI measures — margins, rates, averages
DateTable (Custom)	Time intelligence for year/month trend analysis
Map Visual (Bing Maps)	Geographic sales distribution by country
Power BI Page Navigator	Seamless navigation between Overview and Details

6. Data Preparation & Transformation

The following preprocessing steps were performed using Power Query to ensure clean and analysis-ready data:

- Removal of null, duplicate, and inconsistent records
- Creation of a custom DateTable for time-intelligence calculations
- Standardization of date formats across order and delivery date columns
- Categorization of Delivery Status into defined groups (Late, On-Time, Advance, Cancelled)
- Data type standardization for all numeric and categorical fields
- Definition of DAX measures for all six KPIs

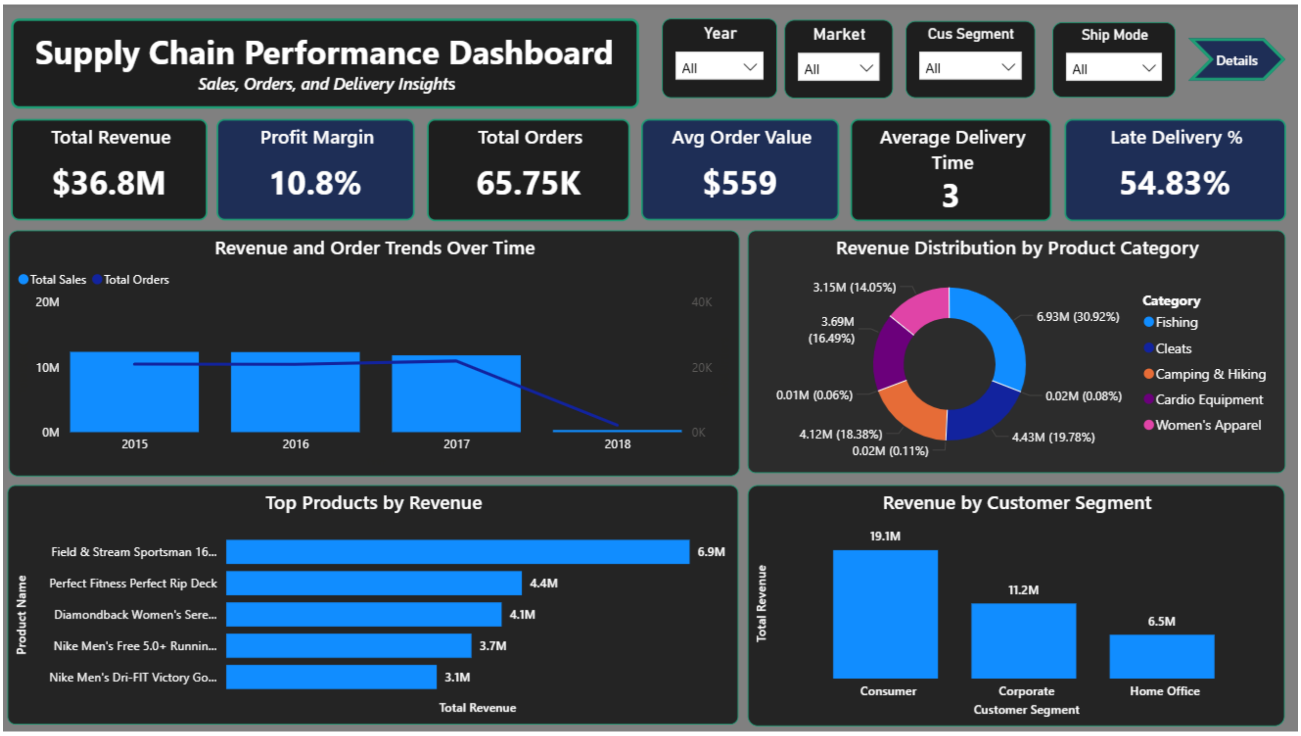
7. Data Modeling Approach

The data model uses a simple relational structure with the main transaction table linked to a custom DateTable for time intelligence. All KPI measures are defined explicitly in a dedicated _Measures table to keep the model clean and reusable.

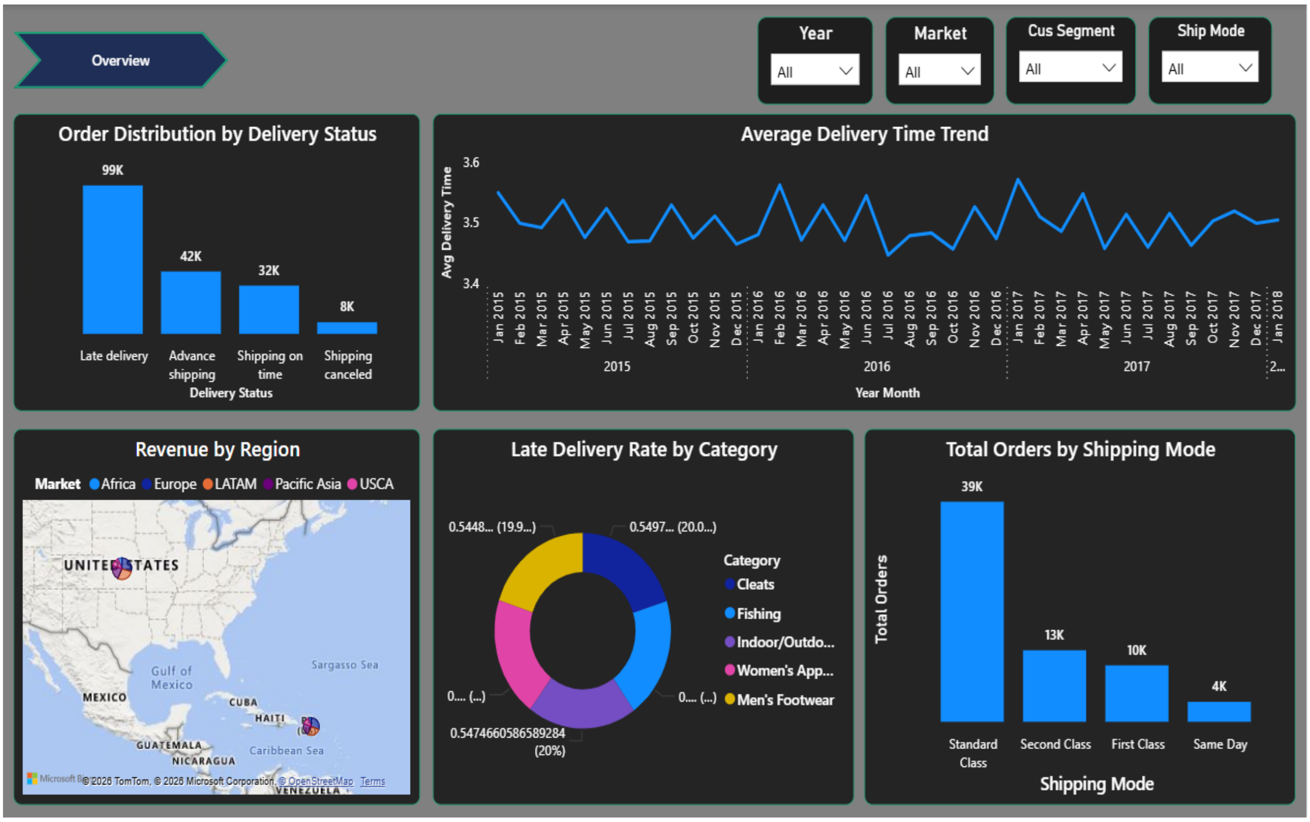
Modeling Decision	Rationale
Main fact table	DataCoSupplyChainDataset holds all transactional records
Custom DateTable	Enables year, month, and quarter-level time filtering
_Measures table	Centralizes all DAX measures for easy maintenance
Implicit cross-filtering	Power BI visuals handle context via slicers and clicks

8.Dashboard Overview

The report consists of two focused pages, each navigable via the built-in page navigator. All slicers are synchronized across both pages for a consistent filtering experience.



Supply Chain Performance Dashboard
Sales, Orders, and Delivery Insights
Year
All
Market
All
Cus Segment
All
Ship Mode
All
Details
Total Revenue
$36.8M
Profit Margin
10.8%
Total Orders
65.75K
Avg Order Value
$559
Average Delivery Time
3
Late Delivery %
54.83%
Revenue and Order Trends Over Time
Total Sales
Total Orders
20M
10M
0M
40K
20K
0K
2015
2016
2017
2018
Revenue Distribution by Product Category
3.15M (14.05%)
3.69M (16.49%)
6.93M (30.92%)
0.01M (0.06%)
0.02M (0.08%)
4.12M (18.38%)
0.02M (0.11%)
4.43M (19.78%)
Category
Fishing
Cleats
Camping & Hiking
Cardio Equipment
Women's Apparel
Top Products by Revenue
Product Name
Field & Stream Sportsman 16...
6.9M
Perfect Fitness Perfect Rip Deck
4.4M
Diamondback Women's Sere...
4.1M
Nike Men's Free 5.0+ Runnin...
3.7M
Nike Men's Dri-FIT Victory Go...
3.1M
Total Revenue
Revenue by Customer Segment
Total Revenue
19.1M
11.2M
6.5M
Consumer
Corporate
Home Office
Customer Segment



Overview
Year
All
Market
All
Cus Segment
All
Ship Mode
All
Order Distribution by Delivery Status
99K
42K
32K
8K
Late delivery
Advance shipping
Shipping on time
Shipping canceled
Delivery Status
Average Delivery Time Trend
Avg Delivery Time
3.6
3.5
3.4
2015
2016
2017
Year Month
Revenue by Region
Market
Africa
Europe
LATAM
Pacific Asia
USCA
UNITED STATES
Gulf of Mexico
Sargasso Sea
MEXICO
CUBA
HAITI
GUATEMALA
Caribbean Sea
NICARAGUA
VENEZUELA
© 2026 TomTom, © 2026 Microsoft Corporation, © OpenStreetMap Terms
Late Delivery Rate by Category
0.5448... (19.9...)
0.5497... (20.0...)
0.... (...)
0.... (...)
0.5474660586589284 (20%)
Category
Cleats
Fishing
Indoor/Outdo...
Women's App...
Men's Footwear
Total Orders by Shipping Mode
Total Orders
39K
13K
10K
4K
Standard Class
Second Class
First Class
Same Day
Shipping Mode

8.1 Page 1 – Overview

Provides a high-level summary of supply chain health across all KPIs and dimensions.

Visual Type	Fields Used	Purpose
KPI Cards (x6)	Total Sales, Orders, Margin %, Avg Order Value,Late Delivery %,	Avg Delivery TimeInstant performance snapshot
Clustered Bar Chart	Product Name vs Total Sales	Identify top revenue-generating products
Line & Column Combo chart	Year, Total Sales, Total Orders	Sales and order volume trends over time
Donut Chart	Category Name vs Total Sales	Revenue share by product category
Column Chart	Customer Segment vs Total Sales	Revenue contribution by segment
Slicers (x4)	Year, Market, Customer Segment, Shipping mode	Dynamic cross-filtering

8.2 Page 2 — Details

Focuses on logistics performance, delivery analysis, and geographic sales distribution.

Visual Type	Fields Used	Purpose
Line Chart	Year, Month, Avg Delivery Time (Days)	Track delivery time trends over time
Donut Chart	Category Name vs Late Delivery %	Categories with highest delay rates
Column Chart	Delivery Status vs Order Count	On-time vs late delivery distribution
Map Visual	Customer Country, Market, Total Sales	Geographic revenue distribution
Column Chart	Shipping Mode vs Total Orders	Order volume by shipping method
Slicers (x4)	Year, Market, Customer Segment, Shipping mode	Consistent cross-page filtering

8.3 Filters & Interactivity

- Four synchronized slicers on each page: Year, Market, Customer Segment, Shipping Mode
- Cross-filtering — clicking any chart element filters all other visuals on the page
- Page Navigator — seamless switching between Overview and Details pages
- Map visual supports zoom and pan for geographic exploration
- Hover tooltips provide granular data values on all chart elements

9.Key Insights

9.1 Delivery Performance

54.83% of all orders experienced late delivery, indicating a significant logistics bottleneck that impacts customer satisfaction. The Fan Shop product category recorded the highest late delivery rate among all categories, making it a priority area for supply chain improvement.

9.2 Shipping Mode Analysis

Standard Class was the most frequently used shipping method, handling approximately 39K of total orders. However, Standard Class also showed the longest average delivery time, suggesting a trade-off between cost efficiency and delivery speed that warrants further review.

9.3 Sales Trends

Sales and order volumes showed a consistent trend across the dataset period, with Total Orders and Total Sales displaying a strong positive correlation across all tracked years. Year-over-year comparisons are available via the Year slicer on the Overview page.

9.4 Customer Segment Performance

The Consumer segment contributed the highest share of total revenue, followed by the Corporate segment. These two segments together account for the majority of sales, making them the primary focus for retention and growth strategies.

9.5 Geographic Distribution

Sales are distributed across multiple global markets including Europe, LATAM, Pacific Asia, USCA, and Africa. The Map visual on the Details page provides an interactive breakdown of Total Sales by Customer Country and Market region.

10.Business Impact

This dashboard delivers tangible value to supply chain and operations teams:

- Reduces time-to-insight by consolidating all KPIs into a single interactive view
- Enables proactive identification of delivery bottlenecks before they escalate
- Supports strategic decisions on shipping mode selection and market prioritization
- Helps segment-level targeting by revealing which customer groups drive revenue
- Provides a reusable template that can be connected to live data sources

11.Conclusion

The Supply Chain Analytics Dashboard successfully transforms raw transactional data into actionable operational insights. It provides a clear understanding of sales performance, delivery efficiency, and logistics patterns across global markets and customer segments.

Such dashboards are essential for modern supply chain management — enabling organizations to shift from reactive problem-solving to proactive, data-driven operations strategy.

12.Future Enhancements

- Predictive delivery delay modeling using Machine Learning (Python / Azure ML)
- Supplier performance and lead time analysis page
- Inventory turnover and stockout rate tracking
- Real-time data integration via API or Azure Synapse connection
- Drill-through pages for per-market or per-category deep dives
- Role-based row-level security (RLS) for market-specific access control